

GLOBAL ACADEMY OF TECHNOLOGY

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Autonomous Institute, Affiliated to VTU

Department of Artificial Intelligence & Machine Learning

AML23702

Advance Natural Language Processing

AML23702_Syllabus.pdf

VASUGI I

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Module-1
Information Extraction

Information Extraction : POS Tagging, Named Entities and Named Entity Tagging, Named Entity Recognition, Relation Extraction, Extracting Events and Time : Temporal Ordering of Events

Textbook : Speech and Language processing: An introduction to Natural Language Processing, Computational Linguistics and speech Recognition by Daniel Jurafsky and James H. Martin

https://web.stanford.edu/~jurafsky/slp3/

Textbook Link:

1. https://web.stanford.edu/~jurafsky/slp3/17.pdf
2. https://web.stanford.edu/~jurafsky/slp3/20.pdf

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Pre-Session Reading

- Short tutorial on POS tagging and named entity recognition, including tagsets and evaluation measures, to support in-class exercises - https://yam.gift/2019/04/09/NLP/SLP/2019-04-09-Ch18-Information-Extraction/
- Survey/overview of event and temporal information extraction to illustrate real-world use cases (news, clinical, legal) - https://www.cfilt.iitb.ac.in/resources/surveys/Temporal_Information_Extraction-Naman-June15.pdf

Sequence Labeling for Part of Speech and Named Entities

- Part of Speech Tagging

Parts of Speech

- From the earliest linguistic traditions (Yaska and Panini 5th C. BCE, Aristotle 4th C. BCE), the idea that words can be classified into grammatical categories
- part of speech, word classes, POS, POS tags
- 8 parts of speech attributed to Dionysius Thrax of Alexandria (c. 1st C. BCE):
- noun, verb, pronoun, preposition, adverb, conjunction, participle, article
- These categories are relevant for NLP today.

Two classes of words: Open vs. Closed

- Closed class words
  - Relatively fixed membership
  - Usually **function** words: short, frequent words with grammatical function
    - determiners: ***a, an, the***
    - pronouns: ***she, he, I***
    - prepositions: ***on, under, over, near, by, …***
- Open class words
  - Usually **content** words: Nouns, Verbs, Adjectives, Adverbs
    - Plus interjections: **oh, ouch, uh-huh, yes, hello**
  - New nouns and verbs like *iPhone* or *to fax*

Open class ("content") words

Nouns

Proper

Janet
Italy

Common

cat, cats
mango

Verbs

Main

eat
went

Auxiliary

can
had

Adjectives *old green tasty*

Adverbs *slowly yesterday*

Numbers

122,312
one

Interjections *Ow hello*

… more

Closed class ("function")

Determiners *the some*

Conjunctions *and or*

Pronouns *they its*

Prepositions *to with*

Particles *off up*

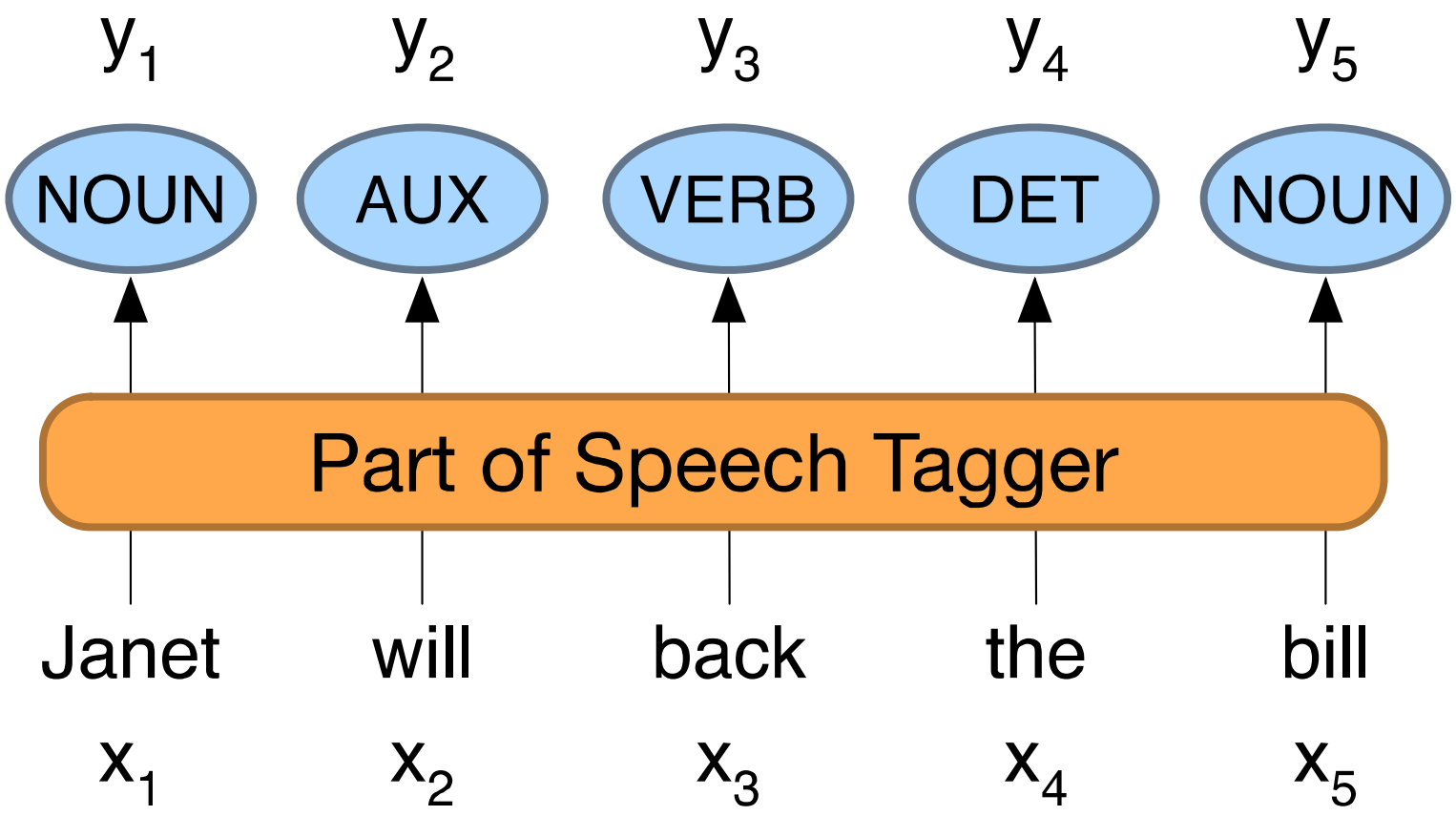
… more

Part-of-Speech Tagging

- Assigning a part-of-speech to each word in a text.
- Words often have more than one POS.
- **book**:
  - VERB: (***Book*** *that flight*)
  - NOUN: (*Hand me that* ***book***).

Part-of-Speech Tagging

Map from sequence $x_1,\ldots,x_n$ of words to $y_1,\ldots,y_n$ of POS tags



"Universal Dependencies" Tagset

Nivre et al. 2016

	Tag	Description	Example
Open Class	**ADJ**	Adjective: noun modifiers describing properties	*red*, *young*, *awesome*
	ADV	Adverb: verb modifiers of time, place, manner	*very*, *slowly*, *home*, *yesterday*
	NOUN	words for persons, places, things, etc.	*algorithm*, *cat*, *mango*, *beauty*
	VERB	words for actions and processes	*draw*, *provide*, *go*
	PROPN	Proper noun: name of a person, organization, place, etc..	*Regina*, *IBM*, *Colorado*
	INTJ	Interjection: exclamation, greeting, yes/no response, etc.	*oh*, *um*, *yes*, *hello*
Closed Class Words	**ADP**	Adposition (Preposition/Postposition): marks a noun's spacial, temporal, or other relation	*in, on, by under*
	AUX	Auxiliary: helping verb marking tense, aspect, mood, etc.,	*can, may, should, are*
	CCONJ	Coordinating Conjunction: joins two phrases/clauses	*and*, *or*, *but*
	DET	Determiner: marks noun phrase properties	*a, an, the, this*
	NUM	Numeral	*one, two, first, second*
	PART	Particle: a preposition-like form used together with a verb	*up, down, on, off, in, out, at, by*
	PRON	Pronoun: a shorthand for referring to an entity or event	*she, who, I, others*
	SCONJ	Subordinating Conjunction: joins a main clause with a subordinate clause such as a sentential complement	*that*, *which*
Other	**PUNCT**	Punctuation	; , ()
	SYM	Symbols like $ or emoji	$, %
	X	Other	asdf, qwfg

Sample "Tagged" English sentences

- There/PRO were/VERB 70/NUM children/NOUN there/ADV ./PUNC
- Preliminary/ADJ findings/NOUN were/AUX reported/VERB in/ADP today/NOUN ’s/PART New/PROPN England/PROPN Journal/PROPN of/ADP Medicine/PROPN

Why Part of Speech Tagging?

- Can be useful for other NLP tasks
  - Parsing: POS tagging can improve syntactic parsing
  - MT: reordering of adjectives and nouns (say from Spanish to English)
  - Sentiment or affective tasks: may want to distinguish adjectives or other POS
  - Text-to-speech (how do we pronounce “lead” or "object"?)
- Or linguistic or language-analytic computational tasks
  - Need to control for POS when studying linguistic change like creation of new words, or meaning shift
  - Or control for POS in measuring meaning similarity or difference

How difficult is POS tagging in English?

- Roughly 15% of word types are ambiguous
- Hence 85% of word types are unambiguous
- *Janet* is always PROPN, *hesitantly* is always ADV
- But those 15% tend to be very common.
- So ~60% of word tokens are ambiguous
- E.g., *back*

  earnings growth took a back/ADJ seat
  a small building in the back/NOUN
  a clear majority of senators back/VERB the bill
  enable the country to buy back/PART debt
  I was twenty-one back/ADV then

POS tagging performance in English

- How many tags are correct? (Tag accuracy)
  - About 97%
    - Hasn't changed in the last 10+ years
    - HMMs, CRFs, BERT perform similarly .
    - Human accuracy about the same
- But baseline is 92%!
  - Baseline is performance of stupidest possible method
    - "Most frequent class baseline" is an important baseline for many tasks
      - Tag every word with its most frequent tag
      - (and tag unknown words as nouns)
  - Partly easy because
    - Many words are unambiguous

Sources of information for POS tagging

Janet will back the bill

AUX/NOUN/VERB? NOUN/VERB?

- Prior probabilities of word/tag
  - "will" is usually an AUX
- Identity of neighboring words
  - "the" means the next word is probably not a verb
- Morphology and wordshape:
  - Prefixes — unable: un- → ADJ
  - Suffixes — importantly: -ly → ADJ
  - Capitalization — Janet: CAP → PROPN

Standard algorithms for POS tagging

- Supervised Machine Learning Algorithms:
- Hidden Markov Models
- Conditional Random Fields (CRF)/ Maximum Entropy Markov Models (MEMM)
- Neural sequence models (RNNs or Transformers)
- Large Language Models (like BERT), finetuned
- All required a hand-labeled training set, all about equal performance (97% on English)
- All make use of information sources we discussed
- Via human created features: HMMs and CRFs
- Via representation learning: Neural LMs

Sequence Labeling for Part of Speech and Named Entities

- Named Entity Recognition (NER)

Named Entities

- **Named entity**, in its core usage, means anything that can be referred to with a proper name. Most common 4 tags:
  - PER (Person): “Marie Curie”
  - LOC (Location): “New York City”
  - ORG (Organization): “Stanford University”
  - GPE (Geo-Political Entity): "Boulder, Colorado"
- Often multi-word phrases
- But the term is also extended to things that aren't entities:
  - dates, times, prices

Named Entity tagging

- The task of named entity recognition (NER):
- find spans of text that constitute proper names
- tag the type of the entity.

NER output

Citing high fuel prices, [ORG **United Airlines**] said [TIME **Friday**] it has increased fares by [MONEY **$6**] per round trip on flights to some cities also served by lower-cost carriers. [ORG **American Airlines**], a unit of [ORG **AMR Corp.**], immediately matched the move, spokesman [PER **Tim Wagner**] said. [ORG **United**], a unit of [ORG **UAL Corp.**], said the increase took effect [TIME **Thursday**] and applies to most routes where it competes against discount carriers, such as [LOC **Chicago**] to [LOC **Dallas**] and [LOC **Denver**] to [LOC **San Francisco**].

Why NER?

- Sentiment analysis: consumer's sentiment toward a particular company or person?
- Question Answering: answer questions about an entity?
- Information Extraction: Extracting facts about entities from text.

Why NER is hard

1) Segmentation
   - In POS tagging, no segmentation problem since each word gets one tag.
   - In NER we have to find and segment the entities!
2) Type ambiguity

[PER Washington] was born into slavery on the farm of James Burroughs.
[ORG Washington] went up 2 games to 1 in the four-game series.
Blair arrived in [LOC Washington] for what may well be his last state visit.
In June, [GPE Washington] passed a primary seatbelt law.

BIO Tagging

- How can we turn this structured problem into a sequence problem like POS tagging, with one label per word?

- [PER Jane Villanueva] of [ORG United] , a unit of [ORG United Airlines Holding] , said the fare applies to the [LOC Chicago] route.

BIO Tagging

- [PER Jane Villanueva] of [ORG United] , a unit of [ORG United Airlines Holding] , said the fare applies to the [LOC Chicago] route.

Words	BIO Label
Jane	B-PER
Villanueva	I-PER
of	O
United	B-ORG
Airlines	I-ORG
Holding	I-ORG
discussed	O
the	O
Chicago	B-LOC
route	O
.	O

How it works

- **B-ENTITY**: marks the first word of an entity.
- **I-ENTITY**: marks the next words in the same entity.
- **O**: marks tokens that are not part of any entity.

Now we have one tag per token!!!

BIO Tagging

- B: token that *begins* a span
- I: tokens *inside* a span
- O: tokens outside of any span

- # of tags (where n is #entity types):
- 1 O tag, for each non- entity type
- *n* B tags,
- *n* I tags
- total of *2n+1*

Words	BIO Label
Jane	B-PER
Villanueva	I-PER
of	O
United	B-ORG
Airlines	I-ORG
Holding	I-ORG
discussed	O
the	O
Chicago	B-LOC
route	O
.	O

BIO Tagging variants: IO and BIOES

- [PER Jane Villanueva] of [ORG United] , a unit of [ORG United Airlines Holding] , said the fare applies to the [LOC Chicago] route.

Words	IO Label	BIO Label	BIOES Label
Jane	I-PER	B-PER	B-PER
Villanueva	I-PER	I-PER	E-PER
of	O	O	O
United	I-ORG	B-ORG	B-ORG
Airlines	I-ORG	I-ORG	I-ORG
Holding	I-ORG	I-ORG	E-ORG
discussed	O	O	O
the	O	O	O
Chicago	I-LOC	B-LOC	S-LOC
route	O	O	O
.	O	O	O

Standard algorithms for NER

- Supervised Machine Learning given a human-labeled training set of text annotated with tags
- Hidden Markov Models
- Conditional Random Fields (CRF)/ Maximum Entropy Markov Models (MEMM)
- Neural sequence models (RNNs or Transformers)
- Large Language Models (like BERT), finetuned

Relation Extraction

Textbook : M1_Information Extraction.pdf (17.2, 17.4)

Relation Extraction

- Relation extraction: finding and classifying semantic relation extraction relations among entities mentioned in a text,
  - like child-of (X is the child-of Y), or part-whole or geospatial relations.
- Relation extraction has close links to populating a relational database, and knowledge graphs
- Datasets of structured relational knowledge graphs knowledge, are a useful way for search engines to present information to users

Relation Extraction

- Up until now we have focused on early stages of the Information Extraction pipeline
  - We have emphasized named entity tagging
- Now we will discuss extracting facts about these entities
  - This can include IS-A facts (similar to named entity types), but also more complicated relations

Extracting relations from text

- Company report: "International Business Machines Corporation (IBM or the company) was incorporated in the State of New York on June 16, 1911, as the Computing-Tabulating-Recording Co. (C-T-R)…"
- Extracted Complex Relation:

  Company-Founding

Company	IBM
Location	New York
Date	June 16, 1911
Original-Name	Computing-Tabulating-Recording Co.

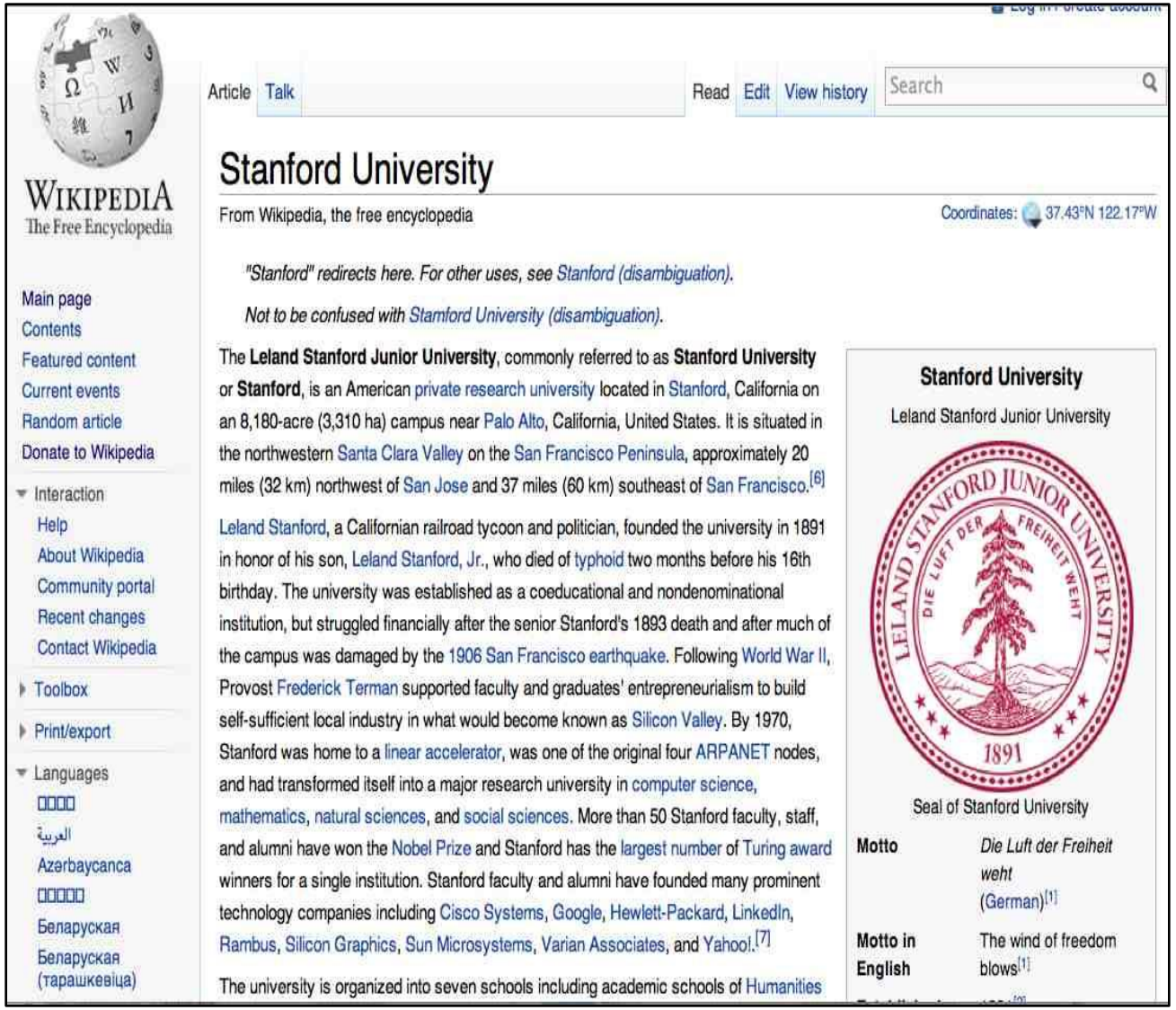
- But we will focus on the simpler task of extracting relation **triples**

  Founding-year(IBM, 1911)

  Founding-location(IBM, New York)

Slide from D. Jurafsky

Extracting Relation Triples from Text



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Stanford University

From Wikipedia, the free encyclopedia

Coordinates: 37.43°N 122.17°W

"Stanford" redirects here. For other uses, see Stanford (disambiguation).

Not to be confused with Stamford University (disambiguation).

The **Leland Stanford Junior University**, commonly referred to as **Stanford University** or **Stanford**, is an American private research university located in Stanford, California on an 8,180-acre (3,310 ha) campus near Palo Alto, California, United States. It is situated in the northwestern Santa Clara Valley on the San Francisco Peninsula, approximately 20 miles (32 km) northwest of San Jose and 37 miles (60 km) southeast of San Francisco.[6]

Leland Stanford, a Californian railroad tycoon and politician, founded the university in 1891 in honor of his son, Leland Stanford, Jr., who died of typhoid two months before his 16th birthday. The university was established as a coeducational and nondenominational institution, but struggled financially after the senior Stanford's 1893 death and after much of the campus was damaged by the 1906 San Francisco earthquake. Following World War II, Provost Frederick Terman supported faculty and graduates' entrepreneurialism to build self-sufficient local industry in what would become known as Silicon Valley. By 1970, Stanford was home to a linear accelerator, was one of the original four ARPANET nodes, and had transformed itself into a major research university in computer science, mathematics, natural sciences, and social sciences. More than 50 Stanford faculty, staff, and alumni have won the Nobel Prize and Stanford has the largest number of Turing award winners for a single institution. Stanford faculty and alumni have founded many prominent technology companies including Cisco Systems, Google, Hewlett-Packard, LinkedIn, Rambus, Silicon Graphics, Sun Microsystems, Varian Associates, and Yahoo![7]

The university is organized into seven schools including academic schools of Humanities

Stanford University
Leland Stanford Junior University

Seal of Stanford University

Motto	*Die Luft der Freiheit weht* (German)[1]
Motto in English	The wind of freedom blows[1]

Main page
Contents
Featured content
Current events
Random article
Donate to Wikipedia
Interaction
Help
About Wikipedia
Community portal
Recent changes
Contact Wikipedia
Toolbox
Print/export
Languages
العربية
Azərbaycanca
Беларуская
Беларуская (тарашкевіца)

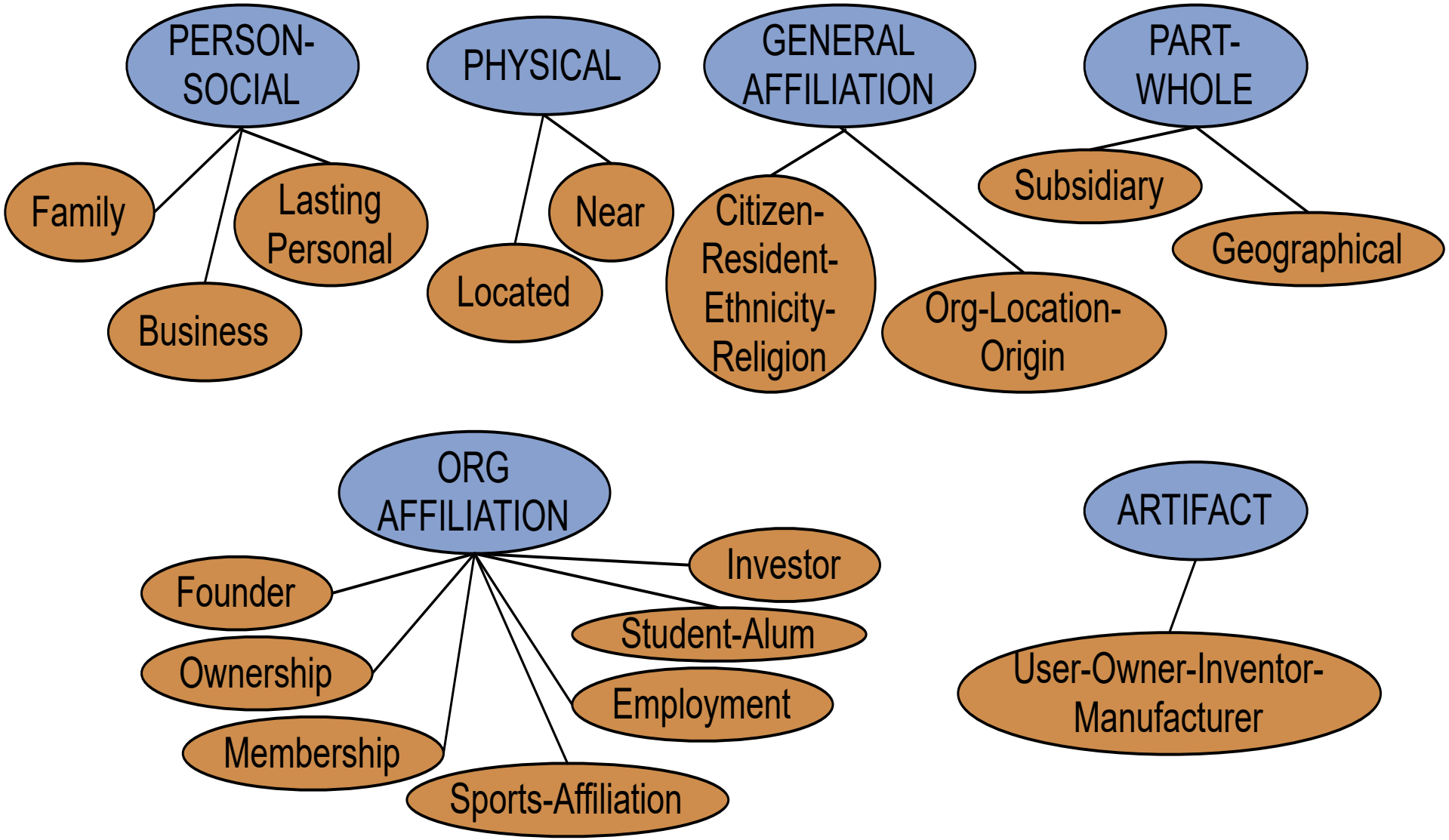
The Leland Stanford Junior University, commonly referred to as Stanford University or Stanford, is an American private research university located in Stanford, California … near Palo Alto, California… Leland Stanford…founded the university in 1891

Stanford EQ Leland Stanford Junior University
Stanford LOC-IN California
Stanford IS-A research university
Stanford LOC-NEAR Palo Alto
Stanford FOUNDED-IN 1891
Stanford FOUNDER Leland Stanford

Slide from D. Jurafsky

Automated Content Extraction (ACE)

17 relations from 2008 "Relation Extraction Task"



Slide from D. Jurafsky

Automated Content Extraction (ACE)

Relations	Types	Examples
Physical-Located	PER-GPE	**He** was in **Tennessee**
Part-Whole-Subsidiary	ORG-ORG	**XYZ**, the parent company of **ABC**
Person-Social-Family	PER-PER	**Yoko**'s husband **John**
Org-AFF-Founder	PER-ORG	**Steve Jobs**, co-founder of **Apple**...

Semantic relations with examples and the named entity types they involve.

Slide from D. Jurafsky

A model-based view of the relations and entities

Citing high fuel prices, [ORG **United Airlines**] said [TIME **Friday**] it has increased fares by [MONEY **$6**] per round trip on flights to some cities also served by lower-cost carriers. [ORG **American Airlines**], a unit of [ORG **AMR Corp.**], immediately matched the move, spokesman [PER **Tim Wagner**] said. [ORG **United**], a unit of [ORG **UAL Corp.**], said the increase took effect [TIME **Thursday**] and applies to most routes where it competes against discount carriers, such as [LOC **Chicago**] to [LOC **Dallas**] and [LOC **Denver**] to [LOC **San Francisco**].

Domain	$\mathscr{D} = \{a,b,c,d,e,f,g,h,i\}$
United, UAL, American Airlines, AMR	a,b,c,d
Tim Wagner	e
Chicago, Dallas, Denver, and San Francisco	f,g,h,i
Classes	
United, UAL, American, and AMR are organizations	$Org = \{a,b,c,d\}$
Tim Wagner is a person	$Pers = \{e\}$
Chicago, Dallas, Denver, and San Francisco are places	$Loc = \{f,g,h,i\}$
Relations	
United is a unit of UAL	$PartOf = \{\langle a,b\rangle, \langle c,d\rangle\}$
American is a unit of AMR	
Tim Wagner works for American Airlines	$OrgAff = \{\langle c,e\rangle\}$
United serves Chicago, Dallas, Denver, and San Francisco	$Serves = \{\langle a,f\rangle, \langle a,g\rangle, \langle a,h\rangle, \langle a,i\rangle\}$

UMLS: Unified Medical Language System

US National Library of Medicine has a network that defines 134 broad subject categories, entity types, and 54 relations between the entities, such as the following:

Entity	Relation	Entity
Injury	disrupts	Physiological Function
Bodily Location	location-of	Biologic Function
Anatomical Structure	part-of	Organism
Pharmacologic Substance	causes	Pathological Function
Pharmacologic Substance	treats	Pathologic Function

https://www.nlm.nih.gov/research/umls/index.html

Slide from D. Jurafsky

Extracting UMLS relations from a sentence

Doppler echocardiography can be used to diagnose left anterior descending artery stenosis in patients with type 2 diabetes



Echocardiography, Doppler DIAGNOSES Acquired stenosis

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Databases of Wikipedia Relations

Wikipedia Infobox

```
{{Infobox university
|image_name= Stanford University seal.svg
|image_size= 210px
|caption = Seal of Stanford University
|name =Stanford University
|native_name =Leland Stanford Junior University
|motto = {{lang|de|"Die Luft der Freiheit weht"}}<br /> ([[German language|German]])<ref
name="casper">{{cite speech|title=Die Luft der Freiheit weht—On and Off|author=Gerhard
Casper|first=Gerhard|last=Casper|authorlink=Gerhard Casper|date=1995-10-
05|url=http://www.stanford.edu/dept/pres-provost/president/speeches/951005dieluft.html}}</ref>
|mottoeng = The wind of freedom blows<ref name="casper" />
|established = 1891<ref>{{cite web |
url=http://www.stanford.edu/home/stanford/history/begin.html | title=Stanford University History |
publisher = Stanford University | accessdate = 2007-04-26}}</ref>
|type = [[private university|Private]]
|calendar= Quarter
|president = [[John L. Hennessy]]
|provost = [[John Etchemendy]]
|city = [[Stanford, California|Stanford]]
|state = California
|country = U.S.
```

Relations extracted from Infobox
Stanford state California
Stanford motto “Die Luft der Freiheit weht”
…

Type	Private
Endowment	US$ 16.5 billion (2011)[3]
President	John L. Hennessy
Provost	John Etchemendy
Academic staff	1,910[4]
Students	15,319
Undergraduates	6,878[5]
Postgraduates	8,441[5]
Location	Stanford, California, U.S.
Campus	Suburban, 8,180 acres (3,310 ha)[6]
Colors	Cardinal red and white

Slide from D. Jurafsky

Relation databases that draw from Wikipedia

- Resource Description Framework (RDF) triples

subject	predicate	object
Golden Gate Park	location	San Francisco
dbpedia:Golden_Gate_Park	dbpedia-owl:location	dbpedia:San_Francisco

- DBPedia: 1 billion RDF triples, 385 from English Wikipedia
- Frequent Freebase relations:

people/person/nationality,
people/person/profession,
biology/organism_higher_classification

location/location/contains
people/person/place-of-birth
film/film/genre

Slide from D. Jurafsky

Patterns for Relation Extraction

- Hand-written rules for relation extraction were used in MUC (such as the Fastus system)
- Recently there has been a renewed wide interest in learning rules for relation extraction focused on precision
  - The presumption is that interesting information occurs many times on the web, with different contexts
    - e.g., how many times does "Barack Obama is the 44th President of the United States" occur on the web?
  - Focusing on high precision is reasonable because the high redundancy will allow us to deal with recall

How to build relation extractors

1. Hand-written patterns
2. Supervised machine learning
3. Semi-supervised and unsupervised
   - Bootstrapping (using seeds)
   - Distant supervision
   - Unsupervised learning from the web

Slide from D. Jurafsky

Rules for extracting IS-A relation

Early intuition from **Hearst (1992)**

- "Agar is a substance prepared from a mixture of **red algae, such as Gelidium,** for laboratory or industrial use"

- What does *Gelidium* mean?
- How do you know?`

Slide from D. Jurafsky

Hearst's Patterns for extracting IS-A relations

(Hearst, 1992): Automatic Acquisition of Hyponyms

"Y such as X ((, X)* (, and|or) X)"
Ex. animals such as cats and dogs

"such Y as X"
"X or other Y"

"X and other Y"
Ex. apples and other fruits

"Y including X"
Ex. fruits including apples, oranges, and bananas

"Y, especially X"
Ex. vehicles, especially cars and trucks

Slide from D. Jurafsky

Hearst's Patterns for extracting IS-A relations

Hearst pattern	Example occurrences
X and other Y	...temples, treasuries, and other important civic buildings.
X or other Y	Bruises, wounds, broken bones or other injuries...
Y such as X	The bow lute, such as the Bambara ndang...
Such Y as X	...such authors as Herrick, Goldsmith, and Shakespeare.
Y including X	...common-law countries, including Canada and England...
Y , especially X	European countries, especially France, England, and Spain...

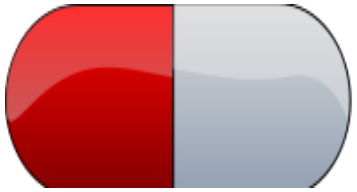
Slide from D. Jurafsky

Extracting Richer Relations Using Rules

- Intuition: relations often hold between specific entities
  - located-in (ORGANIZATION, LOCATION)
  - founded (PERSON, ORGANIZATION)
  - cures (DRUG, DISEASE)
- Start with Named Entity tags to help extract relation!

Slide from D. Jurafsky

Named Entities aren't quite enough.
Which relations hold between 2 entities?



Drug

Cure?

Prevent?

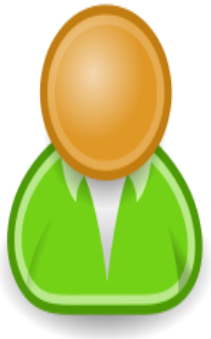
Cause?



Disease

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What relations hold between 2 entities?



PERSON

Founder?

Investor?

Member?

Employee?

President?



ORGANIZATION

Slide from D. Jurafsky

Extracting Richer Relations Using Rules and Named Entities

Who holds what office in what organization?

PERSON, POSITION of ORG

- George Marshall, Secretary of State of the United States

PERSON (named|appointed|chose|*etc.*) PERSON Prep? POSITION

- Truman appointed Marshall Secretary of State

PERSON [be]? (named|appointed|*etc.*) Prep? ORG POSITION

- George Marshall was named US Secretary of State

Slide from D. Jurafsky

Hand-built patterns for relations

- Plus:
  - Human patterns tend to be high-precision
  - Can be tailored to specific domains
- Minus
  - Human patterns are often low-recall
  - A lot of work to think of all possible patterns!
  - Don’t want to have to do this for every relation!
  - We’d like better accuracy

Slide from D. Jurafsky

Supervised machine learning for relations

- Choose a set of relations we'd like to extract
- Choose a set of relevant named entities
- Find and label data
  - Choose a representative corpus
  - Label the named entities in the corpus
  - Hand-label the relations between these entities
  - Break into training, development, and test
- Train a classifier on the training set

```
function FINDRELATIONS(words) returns relations

    relations ← nil
    entities ← FINDENTITIES(words)
    forall entity pairs ⟨e1, e2⟩ in entities do
        if RELATED?(e1, e2)
            relations ← relations+CLASSIFYRELATION(e1, e2)
```

Figure 17.13 Finding and classifying the relations among entities in a text.

Slide from D. Jurafsky

How to do classification in supervised relation extraction

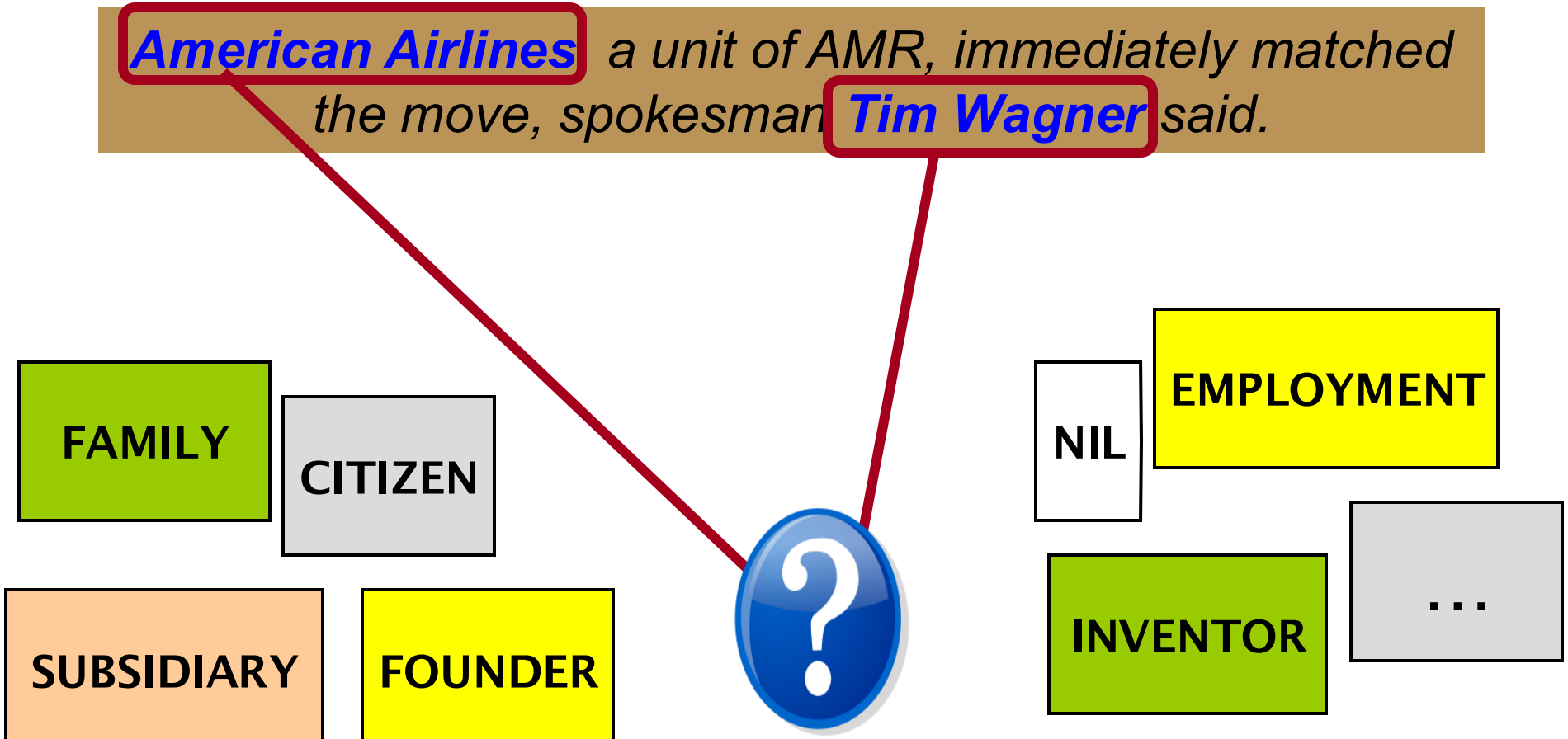
1. Find all pairs of named entities (usually in same sentence)
2. Decide if 2 entities are related
3. If yes, classify the relation

- Why the extra step?
  - Faster classification training by eliminating most pairs
  - Can use distinct feature-sets appropriate for each task.

Slide from D. Jurafsky

Relation Extraction

Classify the relation between two entities in a sentence



Slide from D. Jurafsky

Word Features for Relation Extraction

***American Airlines**, a unit of AMR, immediately matched the move, spokesman **Tim Wagner** said*

Mention 1 Mention 2

- Headwords of M1 and M2, and combination

  Airlines Wagner Airlines-Wagner

- Bag of words and bigrams in M1 and M2

  {American, Airlines, Tim, Wagner, American Airlines, Tim Wagner}

- Words or bigrams in particular positions left and right of M1/M2

  M2: -1 spokesman

  M2: +1 said

- Bag of words or bigrams between the two entities

  {a, AMR, of, immediately, matched, move, spokesman, the, unit}

Slide from D. Jurafsky

Named Entity Type and Mention Level Features for Relation Extraction

***American Airlines**, a unit of AMR, immediately matched the move, spokesman **Tim Wagner** said*

Mention 1 Mention 2

- Named-entity types
  - M1: ORG
  - M2: PERSON
- Concatenation of the two named-entity types
  - ORG-PERSON
- Entity Level of M1 and M2 (NAME, NOMINAL, PRONOUN)
  - M1: NAME [it or he would be PRONOUN]
  - M2: NAME [the company would be NOMINAL]

Slide from D. Jurafsky

Parse Features for Relation Extraction

***American Airlines**, a unit of AMR, immediately matched the move, spokesman **Tim Wagner** said*

Mention 1 Mention 2

- Base syntactic chunk sequence from one to the other

  NP NP PP VP NP NP

- Constituent path through the tree from one to the other

  NP ↑ NP ↑ S ↑ S ↓ NP

- Dependency path

  Airlines matched Wagner said

Slide from D. Jurafsky

Gazetteer and trigger word features for relation extraction

- Trigger list for family: kinship terms
  - parent, wife, husband, grandparent, etc. [from WordNet]
- Gazetteer:
  - Lists of useful geo or geopolitical words
    - Country name list
    - Other sub-entities

Slide from D. Jurafsky

***American Airlines**, a unit of AMR, immediately matched the move, spokesman **Tim Wagner** said.*

Entity-based features		
	Entity_1 type	ORG
	Entity_1 head	*airlines*
	Entity_2 type	PERS
	Entity_2 head	*Wagner*
	Concatenated types	ORGPERS
Word-based features		
	Between-entity bag of words	{ *a, unit, of, AMR, Inc., immediately, matched, the, move, spokesman* }
	Word(s) before Entity_1	NONE
	Word(s) after Entity_2	*said*
Syntactic features		
	Constituent path	$NP \uparrow NP \uparrow S \uparrow S \downarrow NP$
	Base syntactic chunk path	$NP \rightarrow NP \rightarrow PP \rightarrow NP \rightarrow VP \rightarrow NP \rightarrow NP$
	Typed-dependency path	$Airlines \leftarrow_{subj} matched \leftarrow_{comp} said \rightarrow_{subj} Wagner$

Slide from D. Jurafsky

Classifiers for supervised methods

- Now you can use any classifier you like
  - Decision Tree
  - MaxEnt
  - Naïve Bayes
  - SVM
  - …
- Train it on the training set, tune on the dev set, test on the test set

Slide modified from D. Jurafsky

Evaluation of Supervised Relation Extraction

- Compute P/R/F_1 for each relation

$$\mathrm{P} = \frac{\#\text{ of correctly extracted relations}}{\text{Total \# of extracted relations}}$$

$$R = \frac{\#\text{ of correctly extracted relations}}{\text{Total \# of gold relations}}$$

$$F_1 = \frac{2PR}{P+R}$$

Slide from D. Jurafsky

Semi-Supervised Methods

- We'd like to minimize our reliance on having a large training set
- Instead, given a few examples or a few high-precision patterns, we'd like to generalize
  - This is sometimes referred to as "bootstrapping"

Relation Bootstrapping (Hearst 1992)

- Gather a set of seed pairs that have relation R
- Iterate:
  1. Find sentences with these pairs
  2. Look at the context between or around the pair and generalize the context to create patterns
  3. Use the patterns to grep for more pairs

Slide from D. Jurafsky

Bootstrapping

- <Mark Twain, Elmira> Seed tuple
  - Grep (google) for the environments of the seed tuple

  "Mark Twain is buried in Elmira, NY."

  X is buried in Y

  "The grave of Mark Twain is in Elmira"

  The grave of X is in Y

  "Elmira is Mark Twain's final resting place"

  Y is X's final resting place.
- Use those patterns to grep for new tuples
- Iterate

Slide from D. Jurafsky

Dipre: Extract <author, book> pairs

Brin, Sergei. 1998. Extracting Patterns and Relations from the World Wide Web.

- Start with 5 seeds:

Author	Book
Isaac Asimov	The Robots of Dawn
David Brin	Startide Rising
James Gleick	Chaos: Making a New Science
Charles Dickens	Great Expectations
William Shakespeare	The Comedy of Errors

- Find Instances:

  The Comedy of Errors, by William Shakespeare, was

  The Comedy of Errors, by William Shakespeare, is

  The Comedy of Errors, one of William Shakespeare's earliest attempts

  The Comedy of Errors, one of William Shakespeare's most

- Extract patterns (group by middle, take longest common prefix/suffix)

  `?x , by ?y ,` `?x , one of ?y 's`

- Now iterate, finding new seeds that match the pattern

Slide from D. Jurafsky

Distant Supervision

Snow, Jurafsky, Ng. 2005. Learning syntactic patterns for automatic hypernym discovery. NIPS 17
Fei Wu and Daniel S. Weld. 2007. Autonomously Semantifying Wikipeida. CIKM 2007
Mintz, Bills, Snow, Jurafsky. 2009. Distant supervision for relation extraction without labeled data. ACL09

- Combine bootstrapping with supervised learning
  - Instead of 5 seeds,
    - Use a large database to get huge # of seed examples
  - Create lots of features from all these examples
  - Combine in a supervised classifier

Slide from D. Jurafsky

Distant supervision paradigm

- Like supervised classification:
  - Uses a classifier with lots of features
  - Supervised by detailed hand-created knowledge
  - Doesn’t require iteratively expanding patterns
- Like unsupervised classification:
  - Uses very large amounts of unlabeled data
  - Not sensitive to genre issues in training corpus

Slide from D. Jurafsky

Distantly supervised learning of relation extraction patterns

1. For each relation

Born-In

2. For each tuple in big database

<Edwin Hubble, Marshfield>
<Albert Einstein, Ulm>

3. Find sentences in large corpus with both entities

Hubble was born in Marshfield
Einstein, born (1879), Ulm
Hubble's birthplace in Marshfield

4. Extract frequent features (parse, words, etc)

PER was born in LOC
PER, born (XXXX), LOC
PER's birthplace in LOC

5. Train supervised classifier using thousands of patterns

$$P(\text{born-in} \mid f_1, f_2, f_3, \ldots, f_{70000})$$

Slide from D. Jurafsky

Unsupervised relation extraction

M. Banko, M. Cararella, S. Soderland, M. Broadhead, and O. Etzioni. 2007. Open information extraction from the web. IJCAI

- Open Information Extraction:
  - extract relations from the web with no training data, no list of relations

1. Use parsed data to train a "trustworthy tuple" classifier
2. Single-pass extract all relations between NPs, keep if trustworthy
3. Assessor ranks relations based on text redundancy

(FCI, specializes in, software development)

(Tesla, invented, coil transformer)

Slide from D. Jurafsky

Evaluation of Semi-supervised and Unsupervised Relation Extraction

- Since it extracts totally new relations from the web
  - There is no gold set of correct instances of relations!
    - Can't compute precision (don't know which ones are correct)
    - Can't compute recall (don't know which ones were missed)
- Instead, we can approximate precision (only)
  - Draw a random sample of relations from output, check precision manually

$$\hat{P} = \frac{\text{\# of correctly extracted relations in the sample}}{\text{Total \# of extracted relations in the sample}}$$

- Can also compute precision at different levels of recall.
  - Precision for top 1000 new relations, top 10,000 new relations, top 100,000
  - In each case taking a random sample of that set
- But no way to evaluate recall

Slide from D. Jurafsky

Extracting Events and their Times

- The task of event extraction is to identify mentions of events in texts.
- an event mention is any expression denoting an event or state that can be assigned to a particular point, or interval, in time.

[EVENT Citing] high fuel prices, United Airlines [EVENT said] Friday it has [EVENT increased] fares by $6 per round trip on flights to some cities also served by lower-cost carriers. American Airlines, a unit of AMR Corp., immediately [EVENT matched] [EVENT the move], spokesman Tim Wagner [EVENT said]. United, a unit of UAL Corp., [EVENT said] [EVENT the increase] took effect Thursday and [EVENT applies] to most routes where it [EVENT competes] against discount carriers, such as Chicago to Dallas and Denver to San Francisco.

Event Extraction

- Most verbs introduce events.
- Events can be introduced by noun phrases, as in the move and the increase
- Some verbs fail to introduce events,
  - Ex. phrasal verb took effect, which refers to when the event began rather than to the event itself.
  - light verbs such as make, take, and have often fail to denote events;
  - light verbs - event is often expressed by the nominal direct object (took a flight)
- Events can be classified as actions, states, reporting events
  - Ex. (say, report, tell, explain), perception events, and so on

Feature	Explanation
Character affixes	Character-level prefixes and suffixes of target word
Nominalization suffix	Character level suffixes for nominalizations (e.g., *-tion*)
Part of speech	Part of speech of the target word
Light verb	Binary feature indicating that the target is governed by a light verb
Subject syntactic category	Syntactic category of the subject of the sentence
Morphological stem	Stemmed version of the target word
Verb root	Root form of the verb basis for a nominalization
WordNet hypernyms	Hypernym set for the target

Figure 17.24 Features commonly used in both rule-based and machine learning approaches to event detection.

Temporal Ordering of Events

- Events and the temporal expressions in a text having been detected, the next logical task is to use this information to fit the events into a complete timeline.
- A timeline would be useful for applications such as question answering and summarization
- Task is to impose a partial ordering on the events and temporal expressions mentioned in a text.
- Example of such a partial ordering is
  - determination that the fare increase by *American Airlines* came *after* the fare increase by *United* in our sample text

Allen relations

- The temporal relation between events is classified into one of the standard set of Allen relations

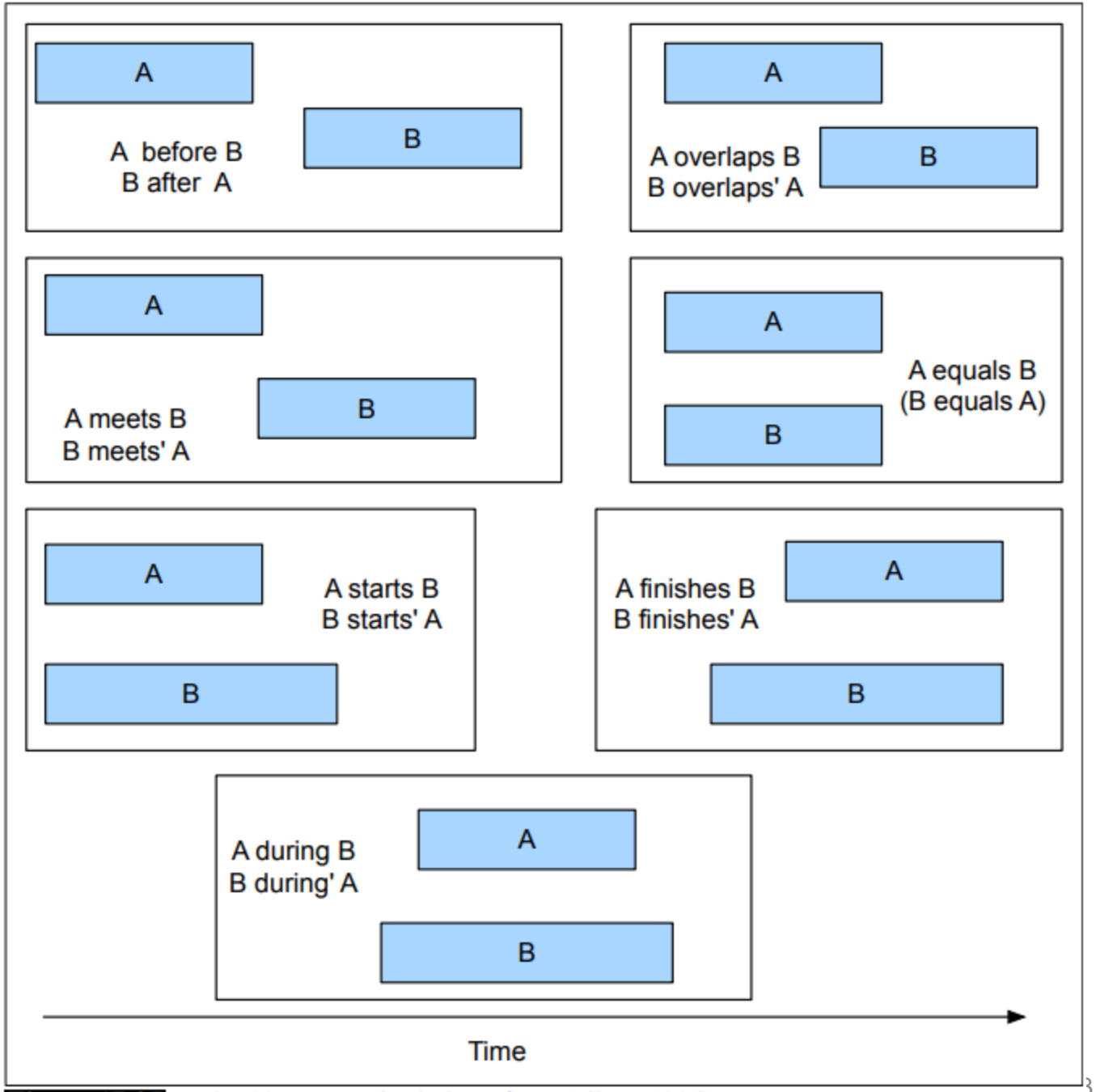


Figure 17.25 The 13 temporal relations from Allen (1984).

Timebank

- The TimeBank corpus consists of annotated text.
- TimeBank 1.2 consists of 183 news articles selected from a variety of sources
- Each article in the TimeBank corpus has had the temporal expressions and event mentions in them explicitly annotated in the TimeML annotation
- TimeML annotation provides temporal links between events and temporal expressions that specify the nature of the relation between them.

```
<TIMEX3 tid="t57" type="DATE" value="1989-10-26"  functionInDocument="CREATION_TIME">
10/26/89  </TIMEX3>

Delta Air Lines earnings <EVENT eid="e1" class="OCCURRENCE"> soared </EVENT> 33% to a
record in  <TIMEX3 tid="t58" type="DATE" value="1989-Q1" anchorTimeID="t57"> the
fiscal first quarter </TIMEX3>, <EVENT eid="e3"  class="OCCURRENCE">bucking</EVENT>
the industry trend toward <EVENT eid="e4" class="OCCURRENCE">declining</EVENT>
profits.
```

Figure 17.26 Example from the TimeBank corpus.